



EXERCISE 2.8

Complex Numbers

Step-by-step solutions to all nine problems — cube roots of unity, De Moivre's theorem, and rotations in the complex plane.



Question 1

If $\omega \neq 1$ is a cube root of unity, show that

$$(a+b\omega+c\omega^2)/(b+c\omega+a\omega^2) + (a+b\omega+c\omega^2)/(c+a\omega+b\omega^2) = -1$$

Step 1 — Key facts about ω

Since ω is a cube root of unity: $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$.

Let $x = a + b\omega + c\omega^2$ (the common numerator).

Step 2 — Rewrite the first denominator

$$\omega^2 \cdot x = \omega^2(a+b\omega+c\omega^2) = a\omega^2 + b\omega^3 + c\omega^4 = a\omega^2 + b + c\omega \quad (\text{using } \omega^3=1, \omega^4=\omega)$$

$$\text{So } b + c\omega + a\omega^2 = \omega^2 x. \rightarrow \text{First fraction} = x / (\omega^2 x) = 1/\omega^2 = \omega$$

Step 3 — Rewrite the second denominator

$$\omega \cdot x = \omega(a+b\omega+c\omega^2) = a\omega + b\omega^2 + c\omega^3 = a\omega + b\omega^2 + c$$

$$\text{So } c + a\omega + b\omega^2 = \omega x. \rightarrow \text{Second fraction} = x / (\omega x) = 1/\omega = \omega^2$$

Step 4 — Add the two fractions

$$\text{Sum} = \omega + \omega^2 = -1 \quad (\text{since } 1 + \omega + \omega^2 = 0)$$

Result: Sum = -1 ✓

Question 2

Show that $(\sqrt{3}/2 + i/2)^5 + (\sqrt{3}/2 - i/2)^5 = -\sqrt{3}$

Step 1 — Write each term in polar (cis) form

$$\sqrt{3}/2 + i/2 = \cos 30^\circ + i \sin 30^\circ = \text{cis}(\pi/6)$$

$$\sqrt{3}/2 - i/2 = \cos 30^\circ - i \sin 30^\circ = \text{cis}(-\pi/6)$$

Step 2 — Apply De Moivre's theorem, raise to the 5th power

$$[\text{cis}(\pi/6)]^5 = \text{cis}(5\pi/6) = \cos 150^\circ + i \sin 150^\circ = -\sqrt{3}/2 + i/2$$

$$[\text{cis}(-\pi/6)]^5 = \text{cis}(-5\pi/6) = \cos(-150^\circ) + i \sin(-150^\circ) = -\sqrt{3}/2 - i/2$$

Step 3 — Add the two results

$$(-\sqrt{3}/2 + i/2) + (-\sqrt{3}/2 - i/2) = -\sqrt{3}/2 - \sqrt{3}/2 = -\sqrt{3}$$

The imaginary parts (+i/2 and -i/2) cancel out exactly.

Result: Sum = $-\sqrt{3}$ ✓



Question 3

Find the value of $[(1 + \sin\pi/10 + i \cos\pi/10) / (1 + \sin\pi/10 - i \cos\pi/10)]^{10}$

Step 1 — Convert sin, cos of $\pi/10$ to cos, sin of a companion angle

Let $\phi = \pi/2 - \pi/10 = 2\pi/5$. Then $\sin(\pi/10) = \cos\phi$ and $\cos(\pi/10) = \sin\phi$.

So numerator = $1 + \cos\phi + i \cdot \sin\phi$, denominator = $1 + \cos\phi - i \cdot \sin\phi$.

Step 2 — Use half-angle identities

$1 + \cos\phi = 2\cos^2(\phi/2)$ and $\sin\phi = 2\sin(\phi/2)\cos(\phi/2)$

Numerator = $2\cos(\phi/2) \cdot \text{cis}(\phi/2)$ | Denominator = $2\cos(\phi/2) \cdot \text{cis}(-\phi/2)$

Step 3 — Simplify the ratio

Ratio = $\text{cis}(\phi/2) / \text{cis}(-\phi/2) = \text{cis}(\phi) = \text{cis}(2\pi/5)$

Step 4 — Raise to the 10th power (De Moivre)

$[\text{cis}(2\pi/5)]^{10} = \text{cis}(10 \times 2\pi/5) = \text{cis}(4\pi) = \cos(4\pi) + i \sin(4\pi) = 1$

Result: Value = 1 ✓

Question 4 — Part (i) & (ii)

Given $2\cos\alpha = x + 1/x$ and $2\cos\beta = y + 1/y$, show the identities below.

Setup — solve for x and y

$$x + 1/x = 2\cos\alpha \Rightarrow x^2 - 2\cos\alpha \cdot x + 1 = 0 \Rightarrow x = \cos\alpha \pm i \sin\alpha$$

Take $x = \operatorname{cis}\alpha$ (so $1/x = \operatorname{cis}(-\alpha)$). Similarly $y = \operatorname{cis}\beta$ (so $1/y = \operatorname{cis}(-\beta)$).

(i) $x/y + y/x = 2\cos(\alpha - \beta)$

$$x/y = \operatorname{cis}\alpha / \operatorname{cis}\beta = \operatorname{cis}(\alpha - \beta) \quad y/x = \operatorname{cis}\beta / \operatorname{cis}\alpha = \operatorname{cis}(\beta - \alpha) = \operatorname{cis}(-(\alpha - \beta))$$

$$x/y + y/x = \operatorname{cis}(\alpha - \beta) + \operatorname{cis}(-(\alpha - \beta)) = 2\cos(\alpha - \beta) \quad \checkmark$$

(ii) $xy - 1/(xy) = 2i \sin(\alpha + \beta)$

$$xy = \operatorname{cis}\alpha \cdot \operatorname{cis}\beta = \operatorname{cis}(\alpha + \beta) \quad 1/(xy) = \operatorname{cis}(-(\alpha + \beta))$$

$$xy - 1/(xy) = \operatorname{cis}(\alpha + \beta) - \operatorname{cis}(-(\alpha + \beta)) = 2i \sin(\alpha + \beta) \quad \checkmark$$



Question 4 — Part (iii) & (iv)

Continuing with $x = \text{cis}\alpha$ and $y = \text{cis}\beta$ from the setup on the previous slide.

(iii) $x^m/y^n - y^n/x^m = 2i \sin(m\alpha - n\beta)$

$$x^m = \text{cis}(m\alpha), \quad y^n = \text{cis}(n\beta)$$

$$x^m/y^n = \text{cis}(m\alpha - n\beta) \quad y^n/x^m = \text{cis}(-(m\alpha - n\beta))$$

$$x^m/y^n - y^n/x^m = \text{cis}(m\alpha - n\beta) - \text{cis}(-(m\alpha - n\beta)) = 2i \sin(m\alpha - n\beta) \quad \checkmark$$

(iv) $x^m y^n + 1/(x^m y^n) = 2\cos(m\alpha + n\beta)$

$$x^m y^n = \text{cis}(m\alpha) \cdot \text{cis}(n\beta) = \text{cis}(m\alpha + n\beta) \quad 1/(x^m y^n) = \text{cis}(-(m\alpha + n\beta))$$

$$x^m y^n + 1/(x^m y^n) = \text{cis}(m\alpha + n\beta) + \text{cis}(-(m\alpha + n\beta)) = 2\cos(m\alpha + n\beta) \quad \checkmark$$

All four identities follow directly from $x = \text{cis}\alpha$, $y = \text{cis}\beta$ $\checkmark$

Question 5

Solve the equation $z^3 + 27 = 0$

Step 1 — Isolate z^3

$$z^3 = -27$$

Step 2 — Write -27 in polar form

$$-27 = 27 \cdot \text{cis}(\pi) \quad (\text{modulus } 27, \text{ argument } \pi + 2k\pi)$$

Step 3 — Take cube roots using De Moivre's theorem

$$z = 27^{1/3} \cdot \text{cis}[(\pi + 2k\pi)/3] = 3 \cdot \text{cis}(\pi/3 + 2k\pi/3), \quad k = 0, 1, 2$$

Step 4 — Substitute $k = 0, 1, 2$

$$k=0: z = 3\text{cis}(\pi/3) = 3(1/2 + i\sqrt{3}/2) = 3/2 + (3\sqrt{3}/2)i$$

$$k=1: z = 3\text{cis}(\pi) = 3(-1 + 0i) = -3$$

$$k=2: z = 3\text{cis}(5\pi/3) = 3(1/2 - i\sqrt{3}/2) = 3/2 - (3\sqrt{3}/2)i$$

Roots: $-3, 3/2 \pm (3\sqrt{3}/2)i$



Question 6

If $\omega \neq 1$ is a cube root of unity, show that the roots of $(z-1)^3 + 8 = 0$ are $-1, 1-2\omega, 1-2\omega^2$

Step 1 — Substitute $w = z - 1$

$$(z-1)^3 + 8 = 0 \Rightarrow w^3 = -8$$

Step 2 — Write -8 using the cube roots of unity

The three cube roots of 1 are $1, \omega, \omega^2$. Since $-8 = (-2)^3$, the cube roots of -8 are:

$$w = -2(1) = -2, \quad w = -2\omega, \quad w = -2\omega^2$$

Check: $(-2\omega)^3 = -8\omega^3 = -8 \cdot 1 = -8$ ✓ (similarly for $-2\omega^2$)

Step 3 — Return to z using $z = w + 1$

$$w = -2 \rightarrow z = -2 + 1 = -1$$

$$w = -2\omega \rightarrow z = 1 - 2\omega$$

$$w = -2\omega^2 \rightarrow z = 1 - 2\omega^2$$

Roots: $-1, 1-2\omega, 1-2\omega^2$ ✓

Question 7

Find the value of $\sum_{k=1}^8 [\cos(2k\pi/9) + i \sin(2k\pi/9)]$

Step 1 — Recognise each term as a 9th root of unity

$$\cos(2k\pi/9) + i \sin(2k\pi/9) = \text{cis}(2k\pi/9) = e^{i2k\pi/9}$$

As k runs $0, 1, 2, \dots, 8$ these are exactly the nine 9th roots of unity.

Step 2 — Use the fact that all n -th roots of unity sum to 0

$$\sum_{k=0}^8 \text{cis}(2k\pi/9) = 0 \quad (\text{standard result for roots of unity})$$

Step 3 — Remove the $k = 0$ term to match the given sum

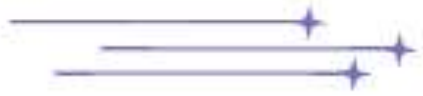
The given sum starts at $k = 1$, so it excludes the $k = 0$ term, which equals $\text{cis}(0) = 1$.

$$\sum_{k=1}^8 \text{cis}(2k\pi/9) = 0 - 1 = -1$$

Result: Sum = -1 ✓



Question 8 — Part (i)



If $\omega \neq 1$ is a cube root of unity, show that $(1-\omega+\omega^2)^6 + (1+\omega-\omega^2)^6 = 128$

Step 1 — Simplify each bracket using $1+\omega+\omega^2 = 0$

$$1 - \omega + \omega^2 = (1+\omega+\omega^2) - 2\omega = 0 - 2\omega = -2\omega$$

$$1 + \omega - \omega^2 = (1+\omega+\omega^2) - 2\omega^2 = 0 - 2\omega^2 = -2\omega^2$$



Step 2 — Raise each to the 6th power

$$(-2\omega)^6 = 64 \cdot \omega^6 = 64 \cdot (\omega^3)^2 = 64 \cdot 1^2 = 64$$

$$(-2\omega^2)^6 = 64 \cdot \omega^{12} = 64 \cdot (\omega^3)^4 = 64 \cdot 1^4 = 64$$

Step 3 — Add the two results

$$(1-\omega+\omega^2)^6 + (1+\omega-\omega^2)^6 = 64 + 64 = 128$$

Result: Sum = 128 ✓



Question 8 — Part (ii)

If $\omega \neq 1$ is a cube root of unity, show that $(1+\omega)(1+\omega^2)(1+\omega^4)(1+\omega^8) \cdots (1+\omega^{2^{11}}) = 1$

Step 1 — Identify the pattern of exponents

The factors are $(1+\omega^{(2^k)})$ for $k = 0, 1, 2, \dots, 11$ — 12 factors in total.

Step 2 — Reduce each exponent mod 3 (since $\omega^3 = 1$)

$2^k \bmod 3$ cycles with period 2: k even $\rightarrow 2^k \equiv 1 \pmod{3}$; k odd $\rightarrow 2^k \equiv 2 \pmod{3}$

So for $k = 0, \dots, 11$ the factor is $(1+\omega)$ when k is even, and $(1+\omega^2)$ when k is odd — 6 of each.

Step 3 — Group the factors

$$\text{Product} = (1+\omega)^6 \cdot (1+\omega^2)^6 = [(1+\omega)(1+\omega^2)]^6$$

Step 4 — Simplify $(1+\omega)(1+\omega^2)$

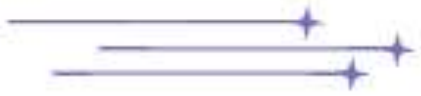
$$(1+\omega)(1+\omega^2) = 1+\omega^2+\omega+\omega^3 = 1+(\omega+\omega^2)+1 = 1+(-1)+1 = 1$$

$$\text{So Product} = 1^6 = 1$$

Result: Product = 1 ✓



Question 9 — Part (i) & (ii)



If $z = 2-2i$, find the rotation of z by ϑ radians (counter-clockwise) about the origin.

Rule: rotating z by θ about the origin gives $z' = z \cdot \text{cis}\theta = z(\cos\theta + i \sin\theta)$

(i) $\theta = \pi/3$



$$\text{cis}(\pi/3) = \cos 60^\circ + i \sin 60^\circ = 1/2 + i\sqrt{3}/2$$

$$z' = (2-2i)(1/2 + i\sqrt{3}/2) = 1 + i\sqrt{3} - i - i^2\sqrt{3} = 1 + i\sqrt{3} - i + \sqrt{3}$$

$$z' = (1+\sqrt{3}) + (\sqrt{3}-1)i$$

(ii) $\theta = 2\pi/3$

$$\text{cis}(2\pi/3) = \cos 120^\circ + i \sin 120^\circ = -1/2 + i\sqrt{3}/2$$

$$z' = (2-2i)(-1/2 + i\sqrt{3}/2) = -1 + i\sqrt{3} + i - i^2\sqrt{3} = -1 + i\sqrt{3} + i + \sqrt{3}$$

$$z' = (\sqrt{3}-1) + (\sqrt{3}+1)i$$



$$(i) \ z' = (1+\sqrt{3}) + (\sqrt{3}-1)i \quad (ii) \ z' = (\sqrt{3}-1) + (\sqrt{3}+1)i$$

Question 9 — Part (iii)

If $z = 2-2i$, find the rotation of z by $\vartheta = 3\pi/2$ radians (counter-clockwise) about the origin.

Step 1 — Recall the rotation rule

$$z' = z \cdot \text{cis}\theta = z(\cos\theta + i \sin\theta)$$

Step 2 — Evaluate $\text{cis}(3\pi/2)$

$$\cos(3\pi/2) = 0, \quad \sin(3\pi/2) = -1 \quad \Rightarrow \quad \text{cis}(3\pi/2) = -i$$

Step 3 — Multiply

$$z' = (2-2i)(-i) = -2i + 2i^2 = -2i - 2$$

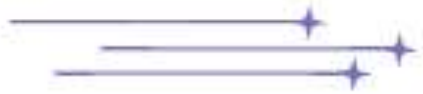
$$z' = -2 - 2i$$

Geometric check

A rotation of $3\pi/2$ (270°) counter-clockwise is the same as 90° clockwise — consistent with $z = 2-2i$ (4th quadrant) landing on $-2-2i$ (3rd quadrant).

Result: $z' = -2 - 2i$ ✓





THANK YOU ✨